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NPN SILICON PHOTOTRANSISTOR LED LAMPS



Lead-Free Parts

LPT2023/A-PF-B01 DATA SHEET

DOC. NO : QW0905-LPT2023/A-PF-B01

REV. : A

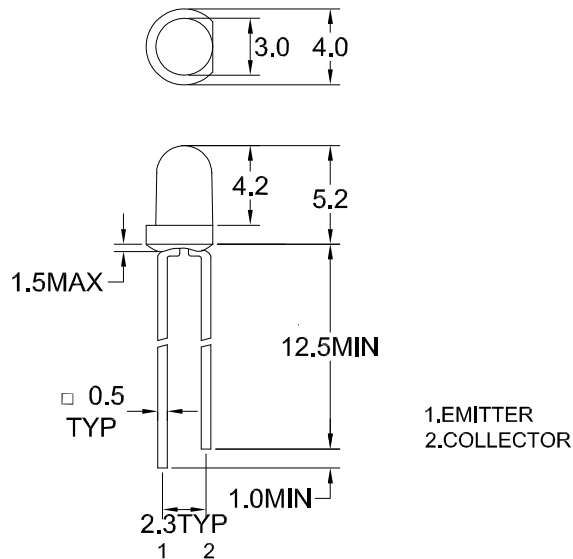
DATE : 23 - Aug. - 2021



PART NO.LPT2023/A-PF-B01

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Package Dimension



Features

- High illumination sensitivity
- Stable characteristics
- Spectrally and mechanically matched with IR emitter

Description

The LPT2023/A-PF-B01 series are silicon nitride passivated NPN planar phototransistors with exceptionally stable characteristics and high illumination sensitivity the cases of LPT2023/A-PF-B01 are encapsulated in water clear plastic T1 3/4 package individualt

Note:1.All dimension are in millimeter tolerance is $\pm 0.25\text{mm}$ unless otherwise noted
2.Specifications are subject to change without notice

• MAXIMUM RATINGS($T_a=25^\circ\text{C}$)

PARAMETER	MAXIMUM RATINGS	UNIT
Power Dissipation	100	mw
Collector-Emitter Voltage	30	V
Emitter-Collector Voltage	5	V
Operating Temperature	-50°C TO $+100^\circ\text{C}$	
Storage Temperature	-50°C TO $+100^\circ\text{C}$	
Lead Soldering Temperature(1.6mm From Body)	260 $^\circ\text{C}$ for 5 seconds	

• ELECTRICAL CHARACTERISTICS($T_a=25^\circ\text{C}$)

PARAMETER	SYMBOL	Min.	Typ.	Max.	UNIT	TEST CONDITION
Collector-Emitter Break down Voltage	$V_{(BR)CEO}$	30			V	$I_c=1\text{mA}$ $E_e=0\text{mw/cm}^2$
Emitter-Collector Break down Voltage	$V_{(BR)ECO}$	5			V	$I_E=100\mu\text{A}$ $E_e=0\text{mw/cm}^2$
Collector-Emitter Saturation Voltage	$V_{CE(sat)}$			0.4	V	$I_c=0.5\text{mA}$ $E_e=20\text{mw/cm}^2$
Rise Time	T_r		5		μS	$V_{CE}=30\text{V}$ $I_C=800\mu\text{A}$, $R_L=1\text{K}\Omega$
Fall Time	T_f		5		μS	
Collector Dark Current	I_{CEO}			100	nA	$V_{CE}=10\text{V}$ $E_e=0\text{mw/cm}^2$
On State Collector Current	$I_{p(on)}$	2		4	mA	$V_{CE}=5\text{v}$ $E_e=1\text{mw/cm}^2$ $\lambda P=940\text{nm}$
		4		8	mA	

Collector Current Classification

BIN	Ion-pt(mA) at 5V	
	Min	Max
I7	2	4
I8	4	8

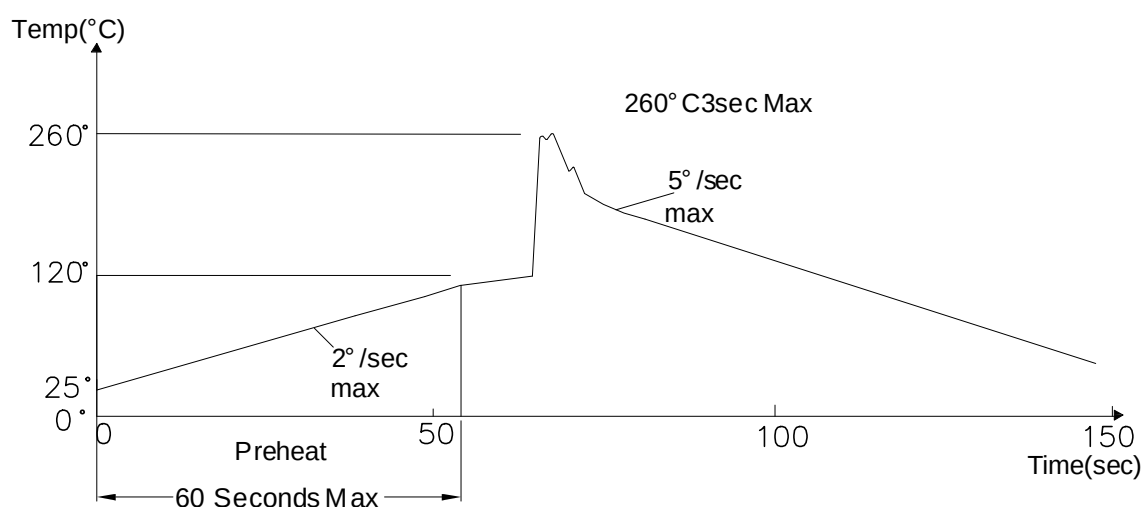
Soldering Condition(Pb-Free)

1.Iron:

Soldering Iron: 30W Max
Temperature 350° C Max
Soldering Time: 3 Seconds Max(One time only)
Distance: 2mm Min(From solder joint to body)

2.Wave Soldering Profile

Dip Soldering
Preheat: 120° C Max
Preheat time: 60seconds Max
Ramp-up
2° C/sec(max)
Ramp-Down:-5° C/sec(max)
Solder Bath: 260° C Max
Dipping Time: 3 seconds Max
Distance: 2mm Min(From solder joint to body)



Note: 1.Wave solder should not be made more than one time.
2.You can just only select one of the soldering conditions as above.

Reliability Test:

Test Item	Test Condition	Description	Reference Standard
Operating Life Test	1.Under Room Temperature 2.If= 20mA 3.t=1000 hrs (-24hrs, +72hrs)	This test is conducted for the purpose of determining the resistance of a part in electrical and thermal stressed.	MIL-STD-750: 1026 MIL-STD-883: 1005 JIS C 7021: B-1
High Temperature Storage Test	1.Ta=85℃±5℃ 2.t=1000 hrs (-24hrs, +72hrs)	The purpose of this is the resistance of the device which is laid under condition of high temperature for hours.	MIL-STD-883:1008 JIS C 7021: B-10
Low Temperature Storage Test	1.Ta=-40℃±5℃ 2.t=1000 hrs (-24hrs, +72hrs)	The purpose of this is the resistance of the device which is laid under condition of low temperature for hours.	JIS C 7021: B-12
High Temperature High Humidity Test	1.Ta=65℃±5℃ 2.RH=90%~95% 3.t=240hrs ±2hrs	The purpose of this test is the resistance of the device under tropical for hours.	MIL-STD-202:103B JIS C 7021: B-11
Thermal Shock Test	1.Ta=105℃±5℃ & -40℃±5℃ (10min) (10min) 2.total 10 cycles	The purpose of this is the resistance of the device to sudden extreme changes in high and low temperature.	MIL-STD-202: 107D MIL-STD-750: 1051 MIL-STD-883: 1011
Solder Resistance Test	1.T.Sol=260℃±5℃ 2.Dwell time=10 ±1sec.	This test is intended to determine the thermal characteristic resistance of the device to sudden exposures at extreme changes in temperature when soldering the lead wire.	MIL-STD-202: 210A MIL-STD-750: 2031 JIS C 7021: A-1
Solderability Test	1.T.Sol=230℃±5℃ 2.Dwell time=5 ±1sec	This test is intended to see soldering well performed or not.	MIL-STD-202: 208D MIL-STD-750: 2026 MIL-STD-883: 2003 JIS C 7021: A-2